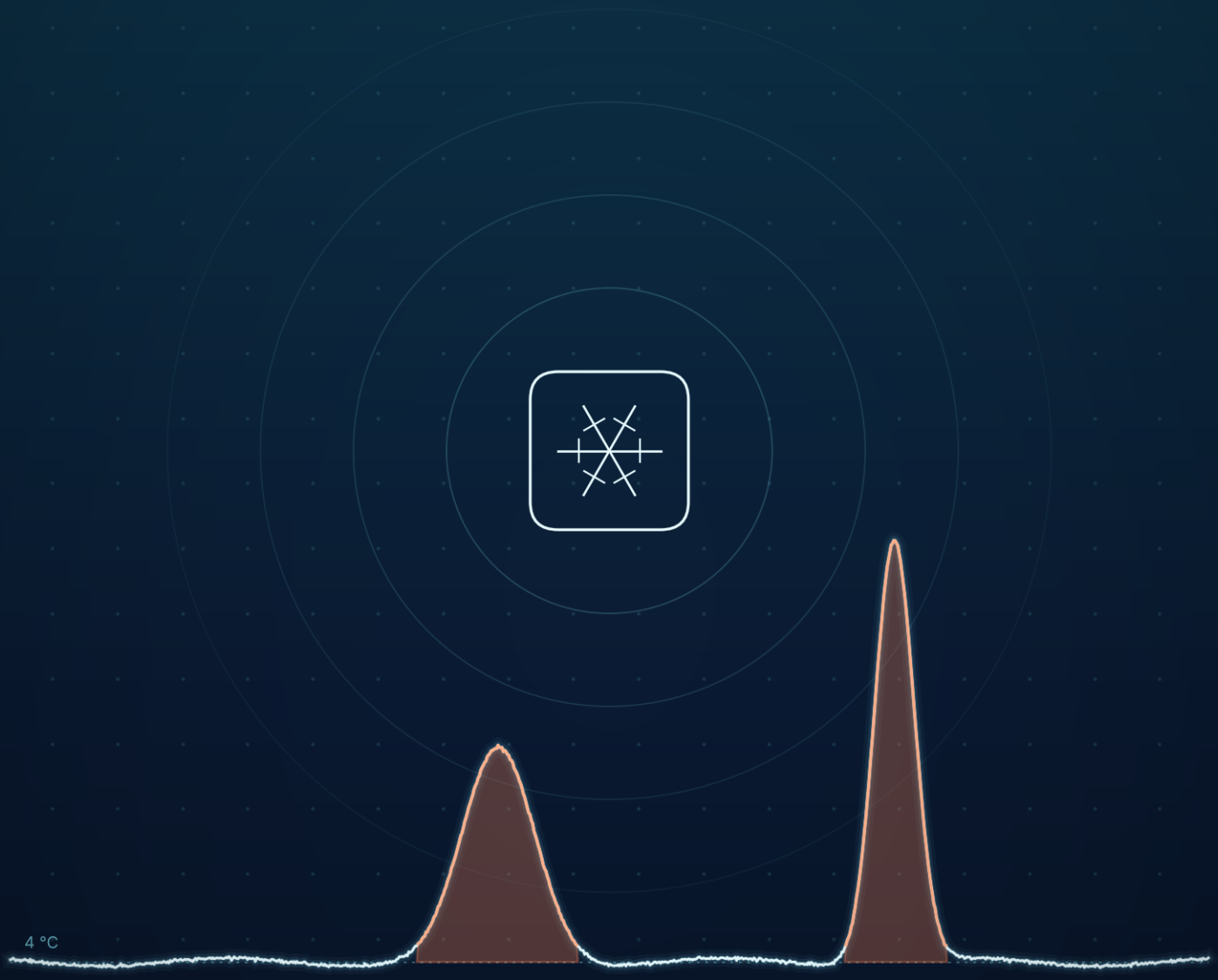


Beyond the 4 °C Alarm

How Much Can Temperature History Tell Us About Spoiling Milk?

by
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A research project in cold-chain monitoring
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Beyond the 4 °C Alarm

There is a quiet moment in every cold chain when nobody is watching. A crate waits on a loading bay. A pouch sits in a shop in an open carton. A delivery bag rides for an hour on a hot afternoon. Nothing dramatic happens, and nothing is recorded. The milk simply gets warmer for a while, and then cold again.

Most of us assume the damage depends on how warm it got. This book asks a more careful question: how much of the damage can a thermometer actually tell us? Bacteria in milk do not read thermometers. They grow at a rate that depends on temperature, but how many there were to begin with, which strains they were, and how the milk was handled before the logger started are all invisible to a temperature probe.

So this is a book about limits as much as about a device. It builds a small, open, tested simulation of a milk cold chain, checks its biology against published experiments and against real records from a public database, and then asks how much a temperature record can and cannot predict. Some of the answer is encouraging. Some of it is humbling. Both parts are the point.

It is written by an engineer who came to the problem from mathematics and computing, and it is meant to be read that way: the models are derived, the code is open, every number can be traced to its source, and the places where the evidence is thin are marked where they occur.

Synopsis

Milk travels through a long chain of hand-offs, and when the chain breaks nobody downstream has a record of what happened. A record of temperature can be turned into an estimate of how much of a milk's shelf life has been used, with growth models that have existed for decades. The open question is how far such an estimate can be trusted, and what limits it.

The question here is narrow and testable. Given a temperature history, how well can an inexpensive on-device calculation tell whether milk will exceed a limit, and what limits that accuracy: the sensor, the sampling interval, the thermal mass of the product, the structure of the growth model, or biology that a thermometer cannot see? A detailed reference growth model plays the role of real milk, and the on-device calculation tries to predict it.

The answer is a loss budget and a set of design rules. An oracle that knows each trip's true biology and temperature sets the ceiling, and information is removed step by step. In every parameter set tried, unobservable biology costs far more accuracy than the sensor or the sampling interval; a plain degree-hour count comes close to the full growth model; and a probability is better calibrated than a single alarm.

The model is tied to data. It reproduces a published fluctuating-temperature experiment to within about 4%. Fitted to 51 real records of natural spoilage bacteria in pasteurised milk it shows that parameters from laboratory isolates overpredict growth by about 1.6 times. Fitted to real records for *Listeria monocytogenes*, a hazard organism, it predicts records it has not seen to within a factor of about 1.4, and the same design rules hold. Chapter 9 states what this does not show.

Chapter Plan

Chapter 1: The Warm Hour (Page 5)
The problem told through the milk itself: what a few warm hours do to a litre of it, why that matters for food waste and for health, and why a temperature log is not the same thing as an answer.

Weight: 10%

Chapter 2: What the Law Actually Says (Page 6)
A close reading of the FSSAI microbiological standards for milk: what the limits mean, what they were written for, and why that decides how a spoilage limit may honestly be defined.

Weight: 8%

Chapter 3: How Milk Spoils (Page 7)
A primer on predictive microbiology, from first principles: growth, lag, temperature, and shelf life as something that accumulates. What earlier work already built, and where the field is stuck.

Weight: 10%

Chapter 4: The Mathematics of a Warm Hour (Page 10)
The model written out and derived: the square-root law, the Baranyi-Roberts reference, the Gompertz shelf-life formula, the accumulator, and the statistics used to judge it. One worked example.

Weight: 12%

Chapter 5: Building the Machine (Page 12)
The computing side: the simulator, the benchmark, the tests, the parameter pipeline and the engineering decisions, including the mistakes that changed the result.

Weight: 15%

Chapter 6: Testing It Against Reality (Page 14)
Where every number comes from, a reproduction of a published experiment, and fits to real ComBase records for natural spoilage flora and for Listeria, with held-out checks.

Weight: 15%

Chapter 7: What a Thermometer Can and Cannot Tell Us (Page 17)
The results: a loss budget separating unobservable biology from sensing and sampling, comparisons against simple rules, and five parameter sets that all tell the same story.

Weight: 20%

Chapter 8: From Model to Machine (Page 21)
What the findings mean for the design of a real monitoring device, as design rules.

Weight: 5%

Chapter 9: Limits, and What Comes Next (Page 22)
What the study does not show, stated plainly, and the experiments that would turn it into a validated result.

Weight: 5%

Chapter 1: The Warm Hour

Take a litre of pasteurised milk that has just left the dairy. It is not sterile. Pasteurisation kills nearly everything that grows in the cold, but a few cells get back in afterwards, from a filler, a pipe, a gasket, a crate. A review of dairy spoilage puts post-pasteurisation contamination, often with *Pseudomonas*, behind about half to three-fifths of the microbial spoilage of fluid milk, and places the onset of quality deterioration near one million cells per millilitre, seven to ten days after pasteurisation [13]. Real loads straight after pasteurisation are tiny, of the order of one to fifty cells in 100 millilitres [13].

Those few cells are on a clock, and the clock is set by temperature. A laboratory study of *Pseudomonas fluorescens* in milk held it at 4 °C and subjected it to two warm periods of four hours at 29 °C. Milk kept cold the whole time reached a million cells per millilitre in about 90 hours. The milk with two warm mornings reached it in about 55 [5]. Eight hours of warmth, spread over two days, took roughly 40% off its life.

This is the warm hour of the title. It is not dramatic. It happens on a loading bay, in a van, in a shop's open carton, and it is invisible to everyone downstream of it. The milk looks the same. It is only the clock inside it that has moved.

Chapter 4 gives the arithmetic of that clock. Here, one number is enough: with the parameters used in this book, a two-hour excursion to 18 °C in an insulated crate uses up about 7% of a shelf life, which is about six hours of refrigerated life, and a simple alarm that checks whether a threshold was crossed cannot say so.

Why it matters to people. Spoiled milk is first of all wasted food and wasted money. But temperature control is also a health matter. The World Health Organization estimated in 2015 that foodborne disease causes about 600 million illnesses and 420,000 deaths a year worldwide, and in its South-East Asia region, which includes India, more than 150 million cases and 175,000 deaths a year [17]. Those deaths come mainly from agents such as norovirus, non-typhoidal *Salmonella* and pathogenic *E. coli*, and the estimates are not specific to milk. This book does not claim that temperature logging prevents any particular number of deaths. It claims something smaller and checkable: that the growth of bacteria in milk responds strongly to temperature, that this includes *Listeria monocytogenes*, a pathogen that grows at refrigerator temperatures (Chapter 6 measures how fast, from real records), and that therefore a record of temperature has value if it can be turned into a sound judgement.

The trouble with a record. A temperature log says how warm the milk was. It does not say how many bacteria there were at the start, which strains they were, or how long they had already been adapting. Two crates with the same log can be at very different places. So the question this book asks is not whether to log temperature. It is how much a log can say once everything else is accounted for.

What is in this book. Chapter 2 reads the Indian standards for milk carefully, because a device that reports a verdict must read them correctly. Chapter 3 explains how milk spoils. Chapter 4 writes the model down and derives it. Chapter 5 describes the software, how it was built and how it was tested. Chapter 6 checks the model against published experiments and against real data. Chapter 7 gives the results. Chapter 8 turns them into design rules for a device, and Chapter 9 says what is still unknown.

*If a reader keeps only three things, they are these. First, **biology is the limit**: what a thermometer cannot see costs far more accuracy than any choice of sensor. Second, **simple rules come close**: a degree-hour count recovers almost everything a full growth model does. Third, **say the uncertainty**: a probability is better calibrated than a yes-or-no alarm.*

Chapter 2: What the Law Actually Says

The FSSAI microbiological standards for milk and milk products are set out in Appendix B of the regulations [1]. They are easy to misread, and a device that claims to report a safe-or-not verdict must read them correctly.

The limit that is often quoted. For pasteurised or boiled milk and flavoured milk, Table 2A gives a three-class plan for aerobic plate count: n = 5 samples, c = 3, m = 3 x 10^4 per mL and M = 5 x 10^4 per mL. Coliforms use n = 5, c = 0 and a limit of less than 10 per mL. The test methods named are IS 5402 / ISO 4833 for plate count (pour plate at 30 °C) and IS 5401 Part 1 / ISO 4832 for coliforms.

What the plan means. In a three-class plan a lot is satisfactory if every value is at or below m. It is acceptable if no more than c values lie between m and M and the rest are at or below m. It is unsatisfactory if any value exceeds M or more than c values exceed m. So 3 x 10^4 and 5 x 10^4 are not interchangeable: one is m, the other is M.

What Table 2A is for. The standard says the process hygiene criteria in Table 2A indicate the acceptable functioning of the production process. They are not to be used as requirements for releasing products to the market, and they apply at the end of the manufacturing process. They are values above which corrective action is required to keep the process hygienic.

What Table 2B is for. The food safety criteria define the acceptability of a batch and apply at the end of manufacture and during shelf life. For pasteurised milk the criteria listed are *Salmonella* and *Listeria monocytogenes*, each absent in 25 mL (n = 5, c = 0). Table 2B lists no *Bacillus cereus* criterion for pasteurised milk.

Sampling and handling. For regulatory sampling the standard requires samples to be stored and transported below 5 °C, not frozen, so that analysis can start within 24 hours. Regulatory testing uses five samples at each of three accredited laboratories, with no retesting.

Consequences for this book. First, the plate-count limit in Table 2A is a process hygiene marker, not an end-of-shelf-life rule. The study therefore evaluates two spoilage limits and names them honestly: scenario A uses 3 x 10^4 CFU/mL as a quality marker, and scenario B uses 10^6 CFU/mL, the level near which a dairy-spoilage review places the onset of quality deterioration [13]. Second, for the hazard organism in Chapter 6 the standard says absent in 25 mL, and a temperature record cannot establish absence. The case study therefore asks only how much *growth* a record implies, with labels of half a log and one log of growth. These are round-number labels and not regulatory limits. Third, any consumer report should say *estimated risk* and never *safe*.

Chapter 3: How Milk Spoils

Predictive microbiology describes how populations of microorganisms change with conditions, using primary models for growth over time, secondary models for how the growth parameters depend on temperature or other factors, and tertiary tools that package these for users [6, 7, 9]. Reviews of the field describe two-step fitting, in which primary and secondary models are fitted in sequence, and one-step fitting, in which they are fitted together [6].

Three phases. A small population in fresh milk does nothing visible at first, a stretch called the lag phase, during which cells adjust. It then grows exponentially, doubling at a steady rate, and finally levels off when it runs out of room or food. Plotted on a logarithmic scale this is an S-shaped curve. For spoilage the interesting part is how long it takes to climb from a handful of cells per millilitre to a million.

The square-root law. A common secondary model for psychrotrophic bacteria in cold storage is the square-root relation of Ratkowsky: the square root of the maximum specific growth rate is proportional to the distance above a notional minimum growth temperature. A University of Tasmania thesis fitted this to 68 growth curves of a fast-growing *Pseudomonas putida* in broth and in milk and found a minimum temperature of -7.7 °C [2]. The same thesis compiled 64 literature values and found the average to be 265.4 K (about -7.75 °C), the same for isolates from dairy and from meat. The consequence is easy to state: growth does not stop at 0 °C. It slows, and each extra degree above the minimum speeds it more than the last.

Shelf life as something that accumulates. If the time to reach the limit at a constant temperature is SL, then a period of time at that temperature uses up the fraction (time / SL) of the shelf life. Adding these fractions over a whole trip gives the cumulative shelf life used. When it reaches 1, the limit has been reached. The calculation is cheap enough for a microcontroller: one table lookup and one addition per reading.

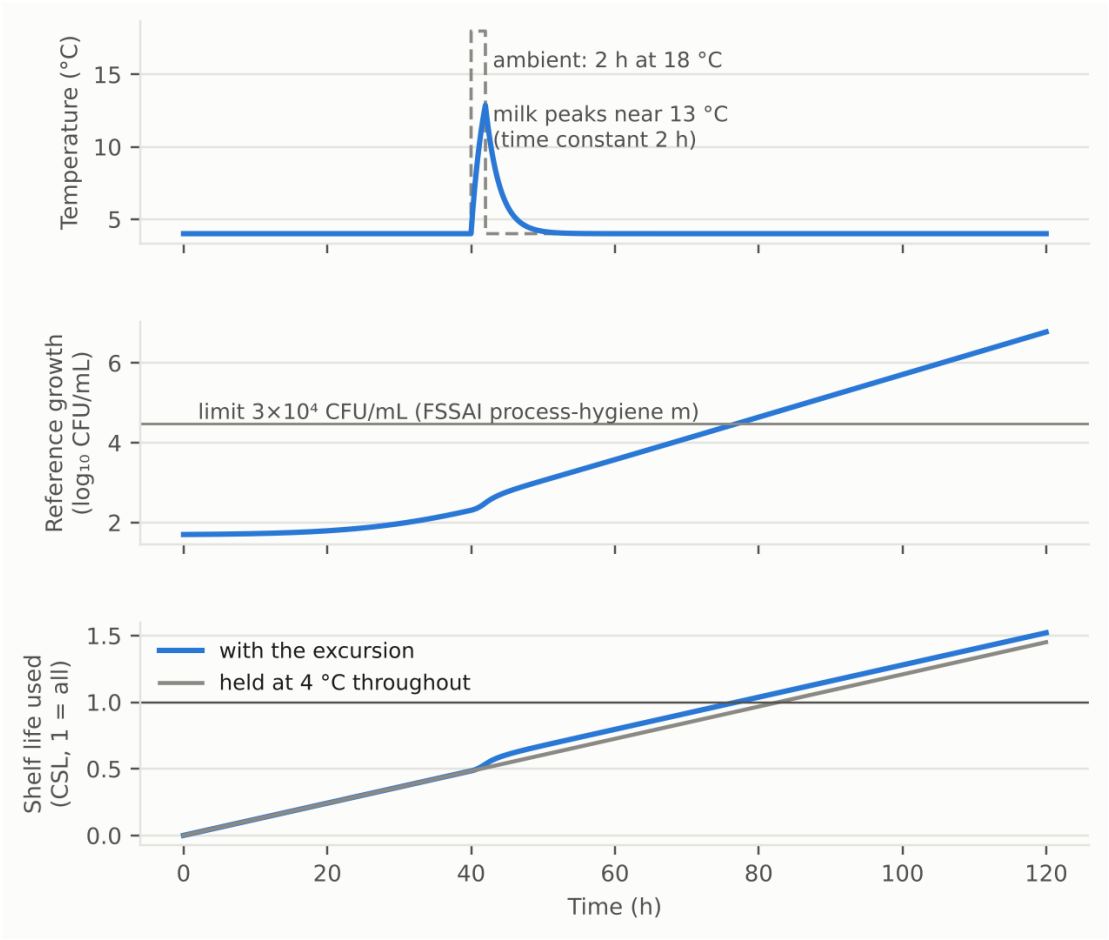


Figure 1. *One trip in the model. Top: ambient and milk temperature. Middle: reference growth (Baranyi) against the 3 x 10^4 CFU/mL marker. Bottom: cumulative shelf life used, with and without the excursion. Starting load is an assumed value.*

What was already built. The idea of turning a temperature log into a spoilage estimate for milk is not new, and this book does not claim it. The Tasmanian thesis [2] developed a milk *Pseudomonas* temperature model, validated it in the laboratory and with industry participants, put it into prototype software, and reviewed time-temperature integrators and loggers. A 2022 poultry study used the same square-root form with fitted limits [15]. A 2016 milk study modelled growth under fluctuating temperature [5], and a 1994 study examined growth after temperature shifts [16]. A 2026 preprint builds a decision framework on temperature streams from cold-chain sensors, combining a simple growth model, a recurrent neural network and a language model, and evaluates it in simulation for pasteurised milk [19]. The closest studies in spirit are uncertainty and sensitivity analyses of shelf-life simulations. A Cornell Monte Carlo model of fluid-milk spoilage found that growth rate and storage temperature move the share of spoiled containers most, ahead of the starting count, with lag the least influential [20]; the same group models spore-former spoilage and has turned it into a supply-chain tool [21, 22]. Another study separated kinetic uncertainty from distribution-temperature variability for a frozen food [23], and a review catalogues how often cold chains break [24]. None of these asks how much a temperature monitor loses to unseen biology, to its sensor, to how often it samples and to its rule, against a known answer. That comparison is what this book is about.

What the recent reviews say is missing. A 2026 review of fresh dairy quality and predictive modelling reports that few studies predict the end of shelf life, that endpoint definition is a major methodological gap, that few models are validated beyond their development data, and that realistic time-temperature testing is uncommon; it recommends prediction intervals or failure probabilities in place of single deterministic estimates [11]. A 2026 review of digital twins for shelf life of fresh-cut produce discusses uncertainty-aware

predictions and calibration but, by its own account, reports no original experiments [12]. Earlier reviews survey models and sensors without testing how much each source of error matters [6-10]. What the sources read for this book do not contain is a measurement of how large each source of error is, on a common footing. That is the gap this book works on. It is a claim about the sources read, and a systematic literature search has not been done.

Chapter 4: The Mathematics of a Warm Hour

This chapter writes the model down. Nothing in it is new mathematics: the pieces come from the predictive-microbiology literature cited in Chapter 3. What is new is putting them in one place, checking each step, and being explicit about which part plays the role of reality and which part plays the role of the device.

4.1 Temperature to growth rate

Let N be the number of cells per millilitre and y = ln N. In the exponential phase, growth is dy/dt = μ, where μ is the specific growth rate in 1/h. The square-root law says μ depends on temperature T as

$$\sqrt{\mu(T)} = b\,(T - T_{\min}), \qquad T > T_{\min}$$

with two constants per organism: a slope b and a notional minimum growth temperature T_{min}. Both are found by linear regression: plot the square root of measured growth rates against temperature and fit a straight line, so that the slope is b and T_{min} = −(intercept)/b. With the Neumeyer parameters b = 0.030 and T_{min} = −7.7 °C [2], growth at the refrigerator temperature of 4 °C is

$$\mu(4\,^{\circ}\mathrm{C}) = (0.030 \times 11.7)^2 \approx 0.123\ \mathrm{h}^{-1}$$

which is a doubling time of ln 2 / 0.123 = 5.6 hours. At 13 °C the same law gives $(0.030 \times 20.7)^2$ = 0.386 per hour, 3.1 times faster. Because the temperature enters squared, a rise from 4 to 13 °C, which is a rise of 9 degrees in a gap of 11.7, more than triples the growth rate.

4.2 The reference model: growth with a memory

Real populations do not start growing at full speed. The Baranyi-Roberts model [14] captures this with a hidden state q that measures how ready the cells are, and growth is throttled by the factor q / (1 + q):

$$\frac{dy}{dt} = \mu(T)\,\frac{q}{1+q}\,(1 - e^{\,y - y_{\max}})$$

$$\frac{dq}{dt} = \mu(T)\,q, \qquad h_0 = \ln\left(1 + \frac{1}{q_0}\right), \qquad \lambda\mu = h_0$$

The state q grows at the same pace as the population, so the lag shortens when the temperature is high and lengthens when it is low, and the product of lag and growth rate stays constant for cells with the same history. That constant, h_0, is the only lag parameter. The last factor in the first equation is a ceiling, y_{max}, the largest count the milk can support. Because μ is evaluated at the current temperature, the model handles any temperature trace with no further work: it is integrated forward one minute at a time in this book. This is the stand-in for real milk.

4.3 The device model: a closed-form shelf life

The device cannot integrate differential equations on every reading, so it uses a different primary model that has a closed form: the modified Gompertz curve,

$$\log_{10} N(t) = \log_{10} N_0 + A \exp\left[-\exp\left(\frac{\mu_{10}\,e}{A}(\lambda - t) + 1\right)\right]$$

where A is the total rise in $\log_{10}$ from start to ceiling, $\mu_{10} = \mu / \ln 10$ is the growth rate per $\log_{10}$ and $\lambda = h_0 / \mu$ is the lag. Setting the left side to the $\log_{10}$ of a limit N_L and solving for t gives the time for the population to reach the limit at constant temperature T:

$$\mathrm{SL}(T) = \frac{h_0}{\mu(T)} + \frac{A}{\mu_{10}(T)\, e}\left[1 - \ln\left(-\ln\frac{L}{A}\right)\right]$$

with $L = \log_{10} N_L - \log_{10} N_0$. This is the shelf life at temperature T. The two models are structurally different on purpose. If the device used the same equations as the reference it would only be copying the truth, and the benchmark would be circular.

4.4 Accumulating shelf life

A reading T_i held for a time Δt_i uses up the fraction Δt_i / $\mathrm{SL}(T_i)$ of the shelf life. Summing over a trip gives the cumulative shelf life used, and the device says the limit has been reached when it reaches 1:

$$\mathrm{CSL} = \sum_i \frac{\Delta t_i}{\mathrm{SL}(T_i)}$$

This is one lookup in a precomputed table and one addition per reading. Why should it be hard to beat with something simpler? Because SL(T) is roughly a constant divided by the growth rate, so

$$\mathrm{SL}(T) \approx \frac{C}{\mu(T)} \;\Rightarrow\; \mathrm{CSL} \approx \frac{1}{C}\int \mu(T(t))\,dt$$

The accumulator is, to a good approximation, the time integral of the growth rate. A degree-hour count above 4 °C is the time integral of $(T - 4)_+$, a first-order stand-in for the same thing, since the growth rate is nearly linear in temperature over a few degrees. Two quantities that are both integrals of increasing functions of temperature will rank most trips the same way. That is the mathematical reason Chapter 7 finds simple rules competitive, and it is also why the comparison is honest about what it can and cannot show.

4.5 A worked example

A crate at 4 °C is exposed to 18 °C air for two hours. The milk has thermal mass, so it warms toward 18 °C with a time constant of two hours and reaches about 13 °C before the air cools again. The growth rate at 13 °C is 3.1 times the rate at 4 °C, so for the part of the excursion spent near 13 °C each hour costs about three hours of refrigerated life, and the warm-up and cool-down add the rest. Integrating the accumulator over the trace gives a cumulative shelf life of 1.52 in place of 1.45 for a trip that stayed at 4 °C: about 7% of one shelf life, which with a label life of 83 hours is about six hours.

4.6 How skill is measured

Each simulated trip is spoiled or not, according to the reference model. Each score the device computes (the accumulator, or a simple rule) is a number per trip, and the question is how well it separates the two groups. The area under the ROC curve answers this without choosing a threshold:

$$\mathrm{AUC} = P(s_{\text{spoiled}} > s_{\text{ok}}) + \frac{1}{2}P(s_{\text{spoiled}} = s_{\text{ok}})$$

It is the probability that a randomly chosen spoiled trip scores higher than a randomly chosen good one. A value of 0.5 means no skill and 1.0 means perfect separation. Confidence intervals come from resampling trips with replacement (the bootstrap). To compare two scores on the same trips the same resampled trips are

used for both, which is the paired bootstrap and is the valid way to test a difference in AUC on shared data.

When the device reports a probability p_k for trip k and the true outcome is o_k (0 or 1), two numbers measure its quality. The Brier score is the mean squared error of the probabilities. The expected calibration error groups trips into bins B_m by predicted probability and averages the gap between the predicted and the observed frequency:

$$\text{Brier} = \frac{1}{n}\sum_{k=1}^{n}(p_k - o_k)^2, \qquad \text{ECE} = \sum_{m}\frac{|B_m|}{n}|\bar{o}_m - \bar{p}_m|$$

A probability that says 70% should be right about 70% of the time. ECE measures how far this is from being true.

Chapter 5: Building the Machine

This chapter is about the software: what was written, how the pieces fit, how they were checked, and which mistakes changed the results. The model is mechanistic, not trained on data in the machine-learning sense. Its parameters are fitted by simple regression to published growth rates, and it is then tested against separate published results and against its own mathematical properties.

5.1 The pieces

Module	What it does
kinetics	Square-root growth rate, closed-form Gompertz shelf life, an interpolated lookup for fast use, and a numerical Baranyi-Roberts integrator. About 100 lines.
simulate	The trip generator (temperature traces with random excursions and thermal inertia), the reference growth along each trace, sensor models and the five scores. Vectorised with NumPy so thousands of trips run at once.
benchmark	Runs all settings, computes AUC with bootstrap intervals, the paired contrasts, and the Monte Carlo probability with Brier score and calibration error. Fixed random seeds.
validate	Fits the model to published rates and to ComBase records, and reproduces the published fluctuating-temperature experiment.
summarize	Condenses every run into the numbers quoted in the paper, the book and the web page, so that no figure is typed by hand.
web	A JavaScript port of the kernel for the companion web page, checked against the Python values (shelf life at 4 °C agrees to three decimals).

5.2 The benchmark, step by step

1. Draw a trip. Base temperature is drawn from 2.5 to 5 °C. Excursions occur at an average of one per 144 hours of horizon, peaking at 12, 14.3, 17.5 or 30.7 °C with weights 0.30, 0.25, 0.30 and 0.15, and last a median of 60 minutes (lognormal, clipped to 10 to 480). Product temperature follows ambient with a first-order time constant: 5 to 20 minutes for a low-mass pouch, 1 to 4 hours for an insulated crate.

2. Grow the truth. Draw the trip's biology: minimum temperature around its mean, a lognormal multiplier on the slope with coefficient of variation 0, 0.10 or 0.25, the lag state from a measured range, and a starting load. Integrate the Baranyi-Roberts model along the trace. A trip is spoiled if the population exceeds the limit within the label horizon.

3. Measure. Add a sensor model to the true temperature: a fixed offset drawn per trip, random noise and quantisation. DS18B20 offsets are within 0.5 °C, PT1000 within 0.15 °C. Sample at 1, 15 or 60 minutes, or

adaptively.

4. Score. Compute the accumulator and four simple rules (degree-hours above 4 °C, time above 5 °C, time above 8 °C, maximum temperature). Compute the oracle too, which uses the trip's true parameters and noise-free temperature.

5. Compare. Compute AUC with bootstrap intervals, paired differences between methods on the same trips, and the loss budget from oracle to real score. For the probability score, draw 80 parameter sets from a prior, run the accumulator under each, and report the fraction that reach the limit.

The label horizon. The horizon is defined the way a printed shelf life is defined in practice: the time the estimator takes to reach the limit at constant 4 °C, rounded up to whole hours. This gives 83 hours for scenario A and 113 hours for scenario B in the baseline. The choice was made from label balance alone, before any method comparison was examined. It makes about half of all trips spoiled by construction, so the results must not be read as real-world spoilage rates.

Model-structure risk. The baseline truth and the device share the assumption that growth follows temperature instantly. Published work shows that real cells can add lag after a temperature shift [5, 16]. A second run therefore re-imposes a quarter of the lag state whenever product temperature departs from, or returns to, its two-hour running mean by more than 4 °C, which is the fraction Zwietering and colleagues found. The device is not told about this.

5.3 How it was tested

Thirteen automated tests run in a few seconds. They check: the Gompertz shelf life against the exponential-growth estimate; agreement of the two growth models in isothermal conditions with matched parameters; the lookup against the closed form; monotonic decrease of shelf life with temperature; the AUC function against a reference implementation and known cases; the paired bootstrap on identical scores (difference exactly zero) and on a real improvement; the probability score against cold and warm histories; the calibration metric on perfect and worst-case inputs; and the reproduction of the published milk experiment within 10%.

5.4 Mistakes that changed the result

Four problems were found, and each changed the numbers. They are listed because a reader should know that the model was corrected by checking it against sources, not accepted on first run.

A horizon that hid the problem. The first run used a fixed 144-hour horizon and found almost no spoiled trips, so every method looked perfect. The horizon was redefined to the printed-shelf-life rule above.

A growth slope that was 3.5 times too low. A placeholder slope of 0.016 never reached a million cells in 140 hours. When two published sources were read, it was replaced by 0.030 and 0.027, which agree. The benchmark was re-run.

A confidence interval read as a spread. The 0.7 K interval on the minimum growth temperature in the thesis is the 95% interval of a mean over 64 sources, not the spread between strains, which is 2.8 K. The simulation now uses an assumed 1.5 K, between the two, and says so.

A level that was wrong for real flora. Parameters from inoculated isolates overpredict growth of natural spoilage flora by about 1.6 times when compared with real ComBase records (Chapter 6). The benchmark was re-run with fitted parameters, and the conclusions held.

5.5 Reproducibility

Every number in the paper and in this book is read from result files by a script. The repository holds the code, the 14 tests, the parameter file with provenance for each value (Appendix A), and one command that regenerates the tables and figures (Appendix B).

Chapter 6: Testing It Against Reality

A simulation is only as believable as its inputs. This chapter records where each biological number comes from, and then asks three questions of real data: does the model reproduce a published experiment, do its parameters describe natural spoilage bacteria, and does the same law describe a hazard organism.

6.1 Two independent sources agree

Neumeyer's thesis [2] gives a square-root slope of 0.00465 for the square root of the reciprocal generation time in minutes, with T_{min} = −7.7 °C, from 68 growth curves. Converted to hours and natural-log growth, this is b = 0.030. Lin and colleagues [5] report growth rates of 0.056, 0.17 and 0.46 per hour at 4, 15 and 29 °C for *P. fluorescens* in milk. Fitting the square-root law to those three values gives b = 0.027 and T_{min} = -9.0 °C.

At 4 °C the Neumeyer parameters imply a generation time of 5.6 hours; Lin and colleagues measured 6.4 hours. Different organisms, different media and different laboratories, and the two agree to within about 15%. The lag state, the product of lag and growth rate, is 3.8 from Lin's lag of 29.5 hours at 4 °C, and ranges from 3.0 to 5.0 across their three temperatures. The simulation draws it from 2.5 to 5.0.

6.2 A published fluctuating-temperature experiment

Lin and colleagues also held milk at 4 °C and subjected it to two warm periods of four hours at 29 °C, the first after 24 hours and the second 24 hours after the first ended. They report that milk at constant 4 °C reached 10^6 CFU/mL in about 90 hours and that the two warm periods cut this to about 55 hours. With parameters fitted only to their growth rates and their lag at 4 °C, the kernel used in this book gives 90 hours at constant 4 °C and 53 hours with the two excursions.

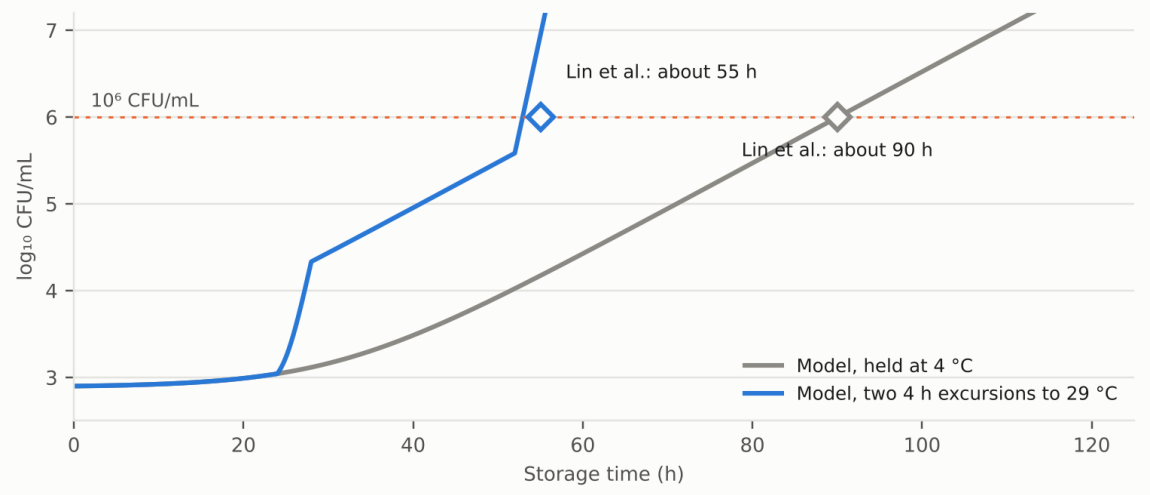


Figure 2. *Reproduction of a published experiment. Lines: this study's Baranyi kernel, with parameters fitted from the published growth rates. Diamonds: the times to 10^6 CFU/mL reported by Lin et al. [5] (about 90 h and about 55 h).*

Quantity	Published	This study	Difference
Time to 10^6 at constant 4 °C (h)	about 90	90	+0%

Quantity	Published	This study	Difference
Time to 10^6 after two 4 h excursions to 29 °C (h)	about 55	53	-4%

The growth rates and the 4 °C lag were used to fit the parameters, so the constant 4 °C result is close by construction. The result after two excursions was not used in fitting. It is the real test, and it is a small one: the published times are approximate values read from the text, not curves, and it is one experiment with one organism in inoculated UHT milk at a starting load of about 800 CFU/mL, far above the load expected after pasteurisation [13].

6.3 Natural spoilage bacteria: ComBase

The studies above inoculated milk with a chosen *Pseudomonas*. What spoils real pasteurised milk is a natural mixture. ComBase, a public database of microbial growth records, holds 51 records of the maximum growth rate of aerobic spoilage bacteria in pasteurised milk at 2.7 to 14.9 °C, from the Tasmanian Institute of Agriculture. The square-root law fits them well (R^2 about 0.9 on the square-root scale), with b = 0.0256 and T_{min} = -6.6 °C [18].

The parameters from the inoculated studies *overpredict* these records. The bias factor is 1.62 and the accuracy factor 1.62, which means predicted growth is about 1.6 times too fast at every temperature. The shape is right and the level is not. Natural spoilage flora in pasteurised milk grows more slowly than an isolate chosen for being fast, which agrees with the thesis choosing its fastest strain as a worst case [2]. The scatter around the fitted curve has a standard deviation of about 16% in the growth rate, a useful empirical anchor for the strain-variability axis used in Chapter 7, though it also contains experimental noise.

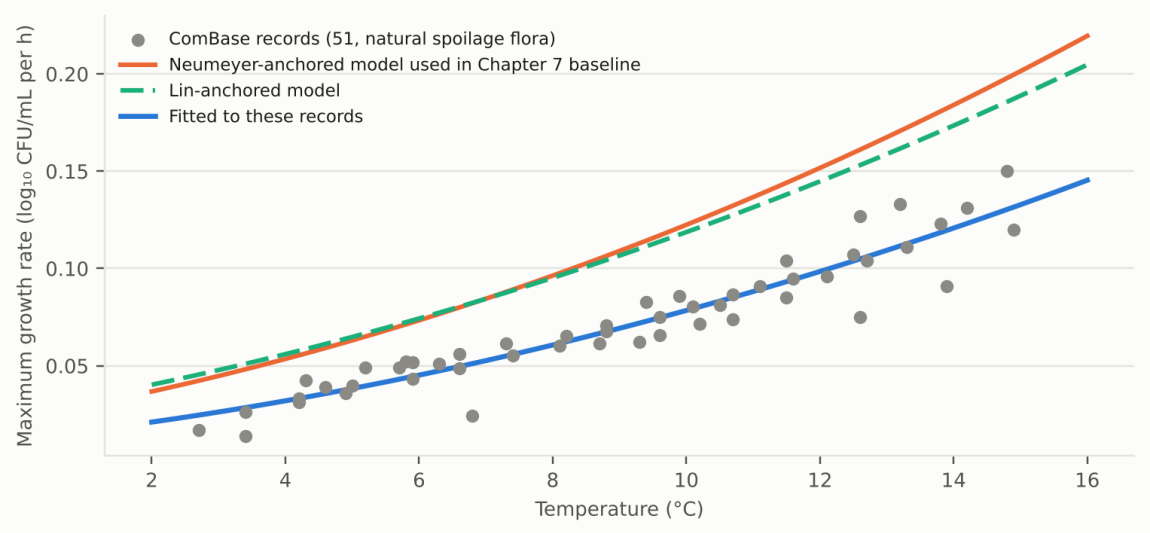


Figure 3. *Maximum growth rate of aerobic spoilage bacteria in pasteurised milk (ComBase records, grey) against the three parameter sets. The Neumeyer- and Lin-anchored curves are about 1.6 times too fast; the fitted curve is in blue.*

6.4 A hazard organism: *Listeria monocytogenes*

Listeria monocytogenes is a pathogen that, unlike most, grows at refrigerator temperatures. The same ComBase search for milk returned records for it. Those with a stated rate at 2 to 16 °C were split by condition. The 46 records with no added factors were used to fit the square-root law, and the 53 records with other stated conditions (sterilised or previously heated milk, added salts or acids) were held out and predicted without refitting.

The fit gives b = 0.0235 (95% bootstrap interval 0.0203 to 0.0265) and T_{min} = -2.3 °C (-4.2 to -0.9 °C). A generation time at 4 °C of 32 hours means growth is slow in a good refrigerator, but it speeds up steeply: at

13 °C it is 5.9 times faster, against about 3.1 times for the spoilage flora (Figure below). On the held-out records the bias factor is 1.02 and the accuracy factor 1.37, so predictions are on average within a factor of about 1.4. The split is by condition and not by study, so some held-out records share a source with fitted ones, and this is a weaker test than a split by study would be.

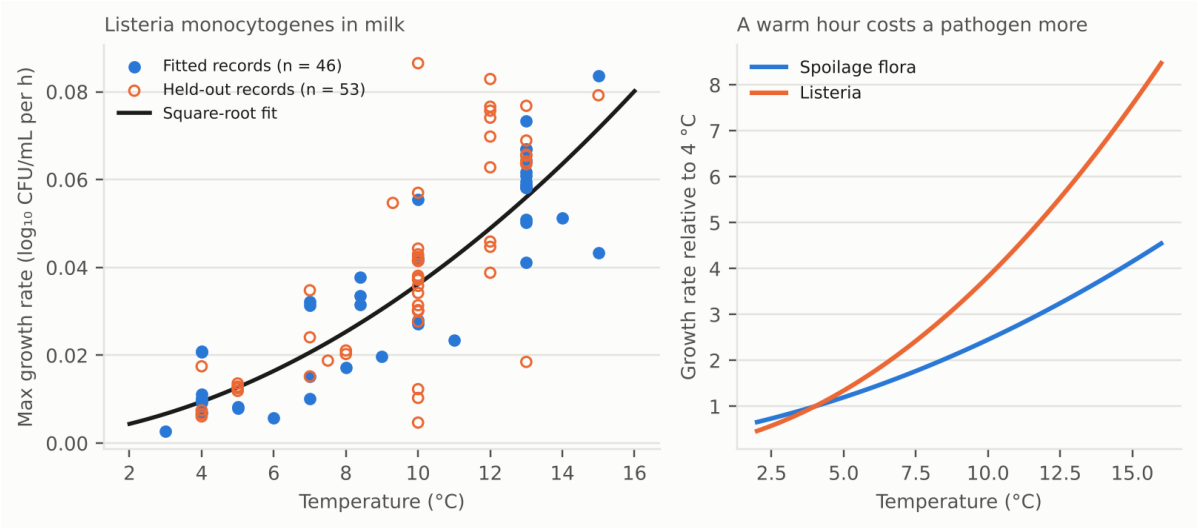


Figure 4. *Listeria monocytogenes in milk. Left: ComBase records, the fit and the held-out records. Right: growth rate relative to 4 °C for the two organism classes. A warm hour costs a pathogen more because its minimum growth temperature is closer to refrigeration.*

6.5 Where the data came from, and how complete it is

The ComBase records were taken from the database's search results for milk. For each record we kept the organism, the temperature, the stated maximum growth rate and the other stated conditions (pH, water activity, added salts or acids, heat treatment).

Of about 480 milk records, 373 were taken, of which 179 state a growth rate; 131 require curve fitting and were not used. Nearly all records are for pathogens, so the spoilage-flora set is 51 records for aerobic spoilage bacteria in pasteurised milk. The fits are consistent with the literature: the *Listeria* minimum temperature of about −2 °C is in the range usually reported for this organism.

Chapter 7: What a Thermometer Can and Cannot Tell Us

All numbers in this chapter are read from the run's output files. The baseline run covers 12 combinations of two spoilage scenarios (label horizons 83 h and 113 h), two product regimes and three levels of strain variability, with 4,000 trips each. Biology is anchored to published sources (Chapter 6), but the strain spread and starting load are assumed (Appendix A). The *pattern* of the results is the finding. Absolute values are not.

1. The loss budget: biology dominates

An oracle that knows each trip's true biology and true temperature separates spoiled from unspoiled trips almost perfectly (AUC about 1.00 by construction). The same kinetic score using nominal biology and an ideal sensor loses between 0.27 and 0.40 AUC across the 12 settings. The loss is 0.28 on average when strains do not differ in growth rate, 0.32 at variability 0.10 and 0.39 at 0.25. Even with no strain variability the loss is large, because the minimum growth temperature, the lag state and the starting load still differ between trips. None of this is in the temperature record, so no sensor and no sampling rate can recover it.

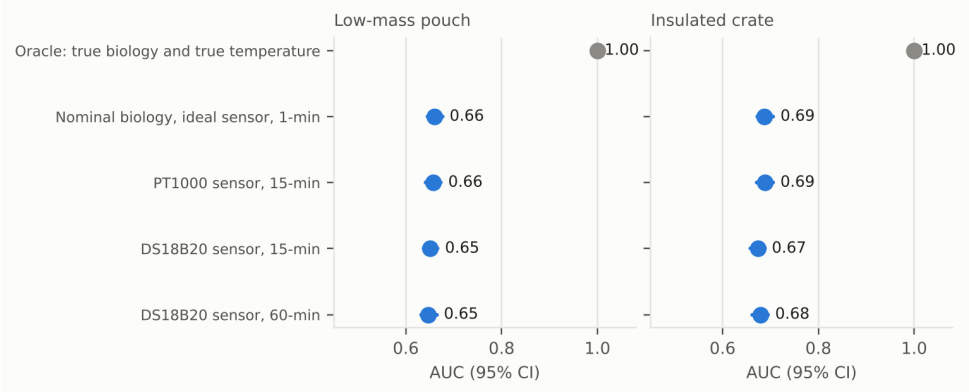


Figure 5. *Loss budget for scenario A at strain variability 0.10. Grey: oracle. Blue: the kinetic score as information is removed. Bars show 95% bootstrap intervals; they are narrow.*

2. Against simple rules: small, consistent gains

On identical trips, the kinetic score beats time above 5 °C in 12 of 12 settings and the maximum temperature in 12 of 12. Against degree-hours above 4 °C, a rule that needs no growth model, the gain is positive in all 12 settings but small: 0.001 to 0.023 AUC, with intervals excluding zero in 10 of 12. It is largest for low-mass pouches at low variability and shrinks as variability rises. Because both scores are close to the time integral of a growth rate, this is expected (Chapters 4 and 9). A practical reading is that most of the value of the kinetic score is available from a simple degree-hour count.

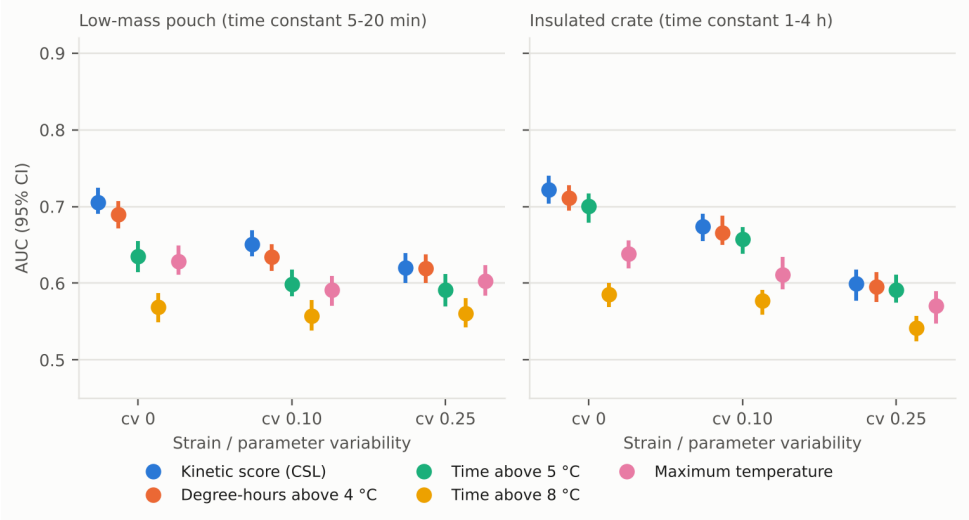


Figure 6. *AUC of five scores from a DS18B20 record sampled every 15 minutes, scenario A. Colours are backed by the table below.*

Regime	Variability	Kinetic	Degree-h	Above 5 °C	Above 8 °C	Max temp
Low-mass	0.00	0.705	0.690	0.635	0.569	0.629
Low-mass	0.10	0.651	0.634	0.599	0.557	0.591
Low-mass	0.25	0.620	0.619	0.591	0.561	0.602
Insulated	0.00	0.722	0.711	0.701	0.586	0.638
Insulated	0.10	0.674	0.666	0.657	0.577	0.611
Insulated	0.25	0.599	0.596	0.592	0.541	0.571

Table: AUC, scenario A, DS18B20 sensor, 15-minute sampling.

3. Sensors, sampling and thermal mass

The accurate PT1000 beats the DS18B20 by 0.003 to 0.015 AUC, with intervals excluding zero in 9 of 12 settings. The effect is small everywhere. Sampling every 60 minutes instead of every minute makes no detectable difference in 11 of 12 settings, and the adaptive rule against fixed one-minute sampling shows an interval excluding zero in 0 of 12. Within this trip generator, thermal mass smooths short excursions enough that the record is not sensitive to how often it is read. The practical message is that a cheap sensor read every hour loses very little, and that effort is better spent on the biology.

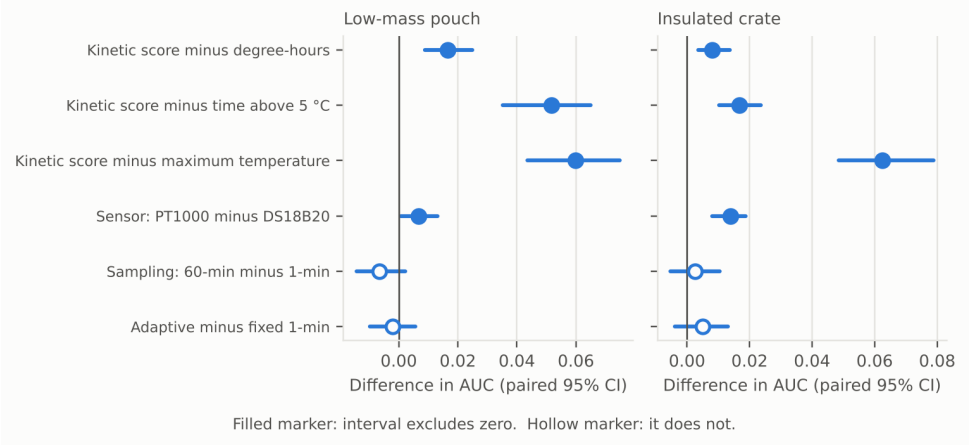


Figure 7. *Paired differences in AUC for scenario A at strain variability 0.10. Filled markers: the interval excludes zero.*

4. Does the conclusion survive a structural error?

In the second run the truth re-imposes a quarter of the lag state after each temperature shift, which the estimator does not model (Chapter 5). The kinetic score still beats degree-hours in all 12 settings (12 of 12 intervals exclude zero), by 0.006 to 0.037. The oracle, which also ignores the shift, falls from an average AUC of 1.000 to 0.988, so the structural error costs about 0.01 AUC at the ceiling. The kinetic score itself loses between 0.01 and 0.04 AUC in scenario A, depending on the setting, as the table shows. The biology gap remains 0.27 to 0.40.

Regime	Variability	Kinetic, baseline	Kinetic, extra lag	Degree-h, baseline	Degree-h, extra lag
Low-mass	0.00	0.705	0.666	0.690	0.630
Low-mass	0.10	0.651	0.619	0.634	0.587
Low-mass	0.25	0.620	0.598	0.619	0.583
Insulated	0.00	0.722	0.711	0.711	0.696
Insulated	0.10	0.674	0.661	0.666	0.651
Insulated	0.25	0.599	0.592	0.596	0.586

Table: AUC, scenario A, DS18B20 sensor, 15-minute sampling, without and with extra lag after temperature shifts.

5. A realistic starting load

The baseline assumes 30 to 100 cells per millilitre at the start, which is high for pasteurised milk [13]. A third run uses 0.1 to 3 cells per millilitre, which stretches the label horizons to 120 h and 154 h. The pattern holds. The kinetic score beats degree-hours by 0.001 to 0.023 AUC (intervals exclude zero in 8 of 12), the biology gap is 0.28 to 0.40, the sensor effect is 0.005 to 0.018, and sampling at 60 minutes shows no detectable loss in 10 of 12 settings.

6. Parameters fitted to real spoilage flora

The ComBase check in Chapter 6 showed natural flora grow about 1.6 times slower than the inoculated-isolate parameters. A fourth run replaces the biology with the fitted values ($b = 0.0256$, $T_{min} = -6.6$ °C, 16% scatter allowed for in the variability axis) and the low starting load. The pattern is unchanged. The kinetic score beats degree-hours by 0.004 to 0.025 AUC (intervals exclude zero in 10 of 12), the biology gap is 0.28 to 0.40, the sensor effect is 0.002 to 0.015, and sampling at 60 minutes shows no detectable loss in 12 of 12 settings. The conclusions do not depend on which parameter set is used.

7. A hazard organism: Listeria

The fifth run replaces the spoilage flora with *Listeria monocytogenes*, using the parameters fitted to real ComBase records in Chapter 6 ($b = 0.0235$, $T_{min} = -2.3$ °C). A trip is positive if the pathogen grows by half a log (scenario A, label horizon 100 h) or by one log (scenario B, 202 h) over its starting load. These are round-number labels, not regulatory limits, and the lag and starting load are assumed. The pattern is the same. The kinetic score beats degree-hours by 0.002 to 0.023 AUC (intervals exclude zero in 8 of 12), the biology gap is 0.09 to 0.34, the sensor effect is 0.005 to 0.020, and sampling hourly shows no detectable loss in 12 of 12 settings. One difference matters: for the half-log label (scenario A) even the oracle reaches only 0.88 to 0.92 AUC, because a limit that close to the start lies where the closed-form Gompertz shelf life is a poor approximation of the reference. Label choice interacts with the estimator's structure, and the later calibration section reports the consequence. The design rules of Chapter 8 therefore apply to a hazard organism as well as to spoilage, with the caveat that these results say nothing about illness risk.

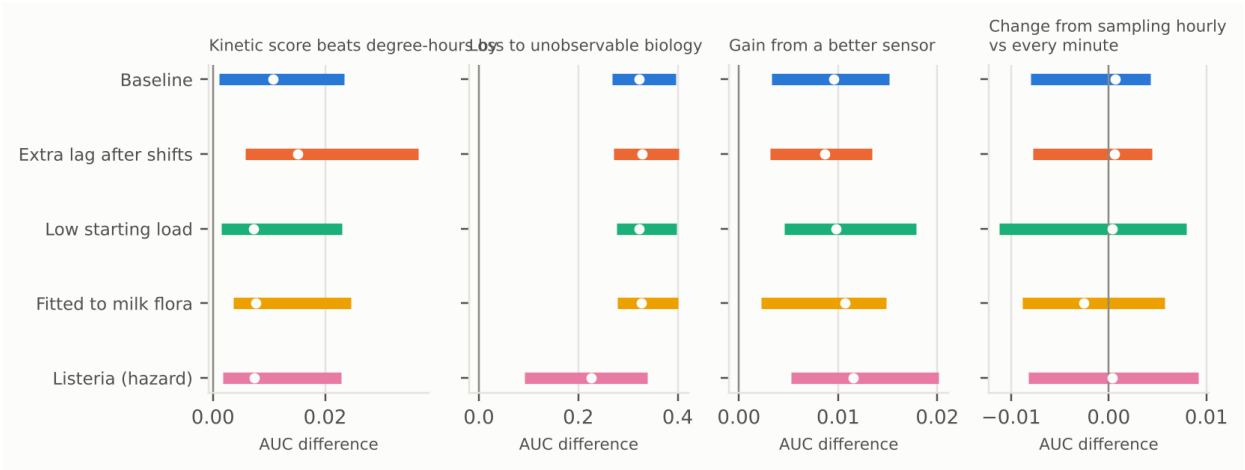


Figure 8. *The same four effects across all five runs. Each bar spans the 12 settings of a run (dot: median), as AUC differences on identical trips. The pattern does not depend on the parameter set.*

8. An uncertainty-aware score is better calibrated

Reporting the fraction of 80 plausible parameter sets that reach the limit gives a probability. With a prior matching the truth, its calibration error is 0.03 to 0.07, against 0.33 to 0.43 for a single alarm using nominal values, and its Brier score is lower in all 12 settings. If the prior is overconfident about the growth rate, calibration error rises to as much as 0.13 when variability is 0.25 but stays well below the single alarm. The result needs careful reading: a single alarm is a hard 0 or 1, so it is penalised for confident errors by construction, and the comparison shows the value of expressing uncertainty, not a better model. The matched-prior case is an upper bound, because real priors will never match exactly. For the *Listeria* run the probability still beats the single alarm in every setting, but with the half-log label its calibration is poor in absolute terms, for the structural reason given above.

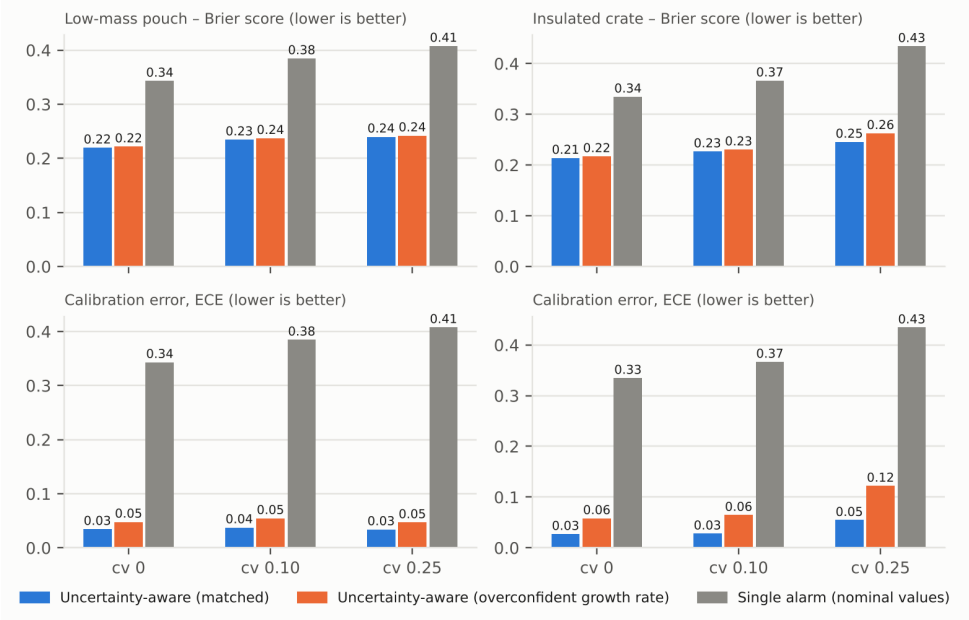


Figure 9. *Brier score and expected calibration error for scenario A. Lower is better. Values are printed on the bars.*

9. What the results do and do not support

Supported within the simulation: unobservable biology is the largest loss by a wide margin; a kinetic score offers a small but consistent edge over a degree-hour count, even when the estimator ignores extra lag after temperature shifts; sensor accuracy matters little; sampling interval does not matter at these settings; and a probability with stated uncertainty is better calibrated than a single alarm. Not supported: any claim about real milk, any claim that the strain spread and starting load assumed here are right, or any claim about illness risk.

Chapter 8: From Model to Machine

The findings in Chapter 7 are about any temperature record, whether it comes from a commercial data logger or a stick-on probe. Temperature loggers and time-temperature indicators are already products; this book does not survey them. What the study adds is a ranking of what matters when one is designed, and the ranking is not the one a hardware project would guess.

Rule 1: spend little on the thermometer. A digital DS18B20 probe, specified at plus or minus 0.5 °C, is within about 0.01 to 0.02 AUC of a PT1000 read through a MAX31865 converter, which is specified at about plus or minus 0.15 °C. A better probe is a poor way to improve a spoilage estimate.

Rule 2: sample every 15 to 60 minutes. Under the assumed excursion model, hourly sampling is indistinguishable from every-minute sampling in most settings. An adaptive rule that samples faster when the product is warm or changing gave no detectable advantage either. The caveat is the excursion model: if short, sharp excursions are common, this would change.

Rule 3: compute degree-hours first, add the growth model if cheap. A degree-hour count above 4 °C recovers nearly all the skill of the accumulator. The accumulator costs one lookup per reading, so it is nearly free on a microcontroller, and it adds a small, consistent edge, larger when extra lag after temperature shifts is present.

Rule 4: spend effort on the biology. The largest loss is what a thermometer cannot see: starting load, strain, lag. A design that records handling history, supports a cheap load or hygiene proxy, or lets a supervisor enter a batch's age could recover part of this. No sensor upgrade can.

Rule 5: report a probability, and never the word 'safe'. A single yes-or-no alarm was poorly calibrated in every run. A probability from a set of plausible parameters was much better. For the hazard organism the standard asks for absence, which a temperature record cannot establish, so a report should say *estimated risk*.

Chapter 9: Limits, and What Comes Next

These limits are listed so a reader, a reviewer or a co-author can find them without searching.

Partly assumed biology. The growth slope, minimum temperature and lag state are sourced or fitted. The strain spread, the starting load, the carrying capacity, the excursion pattern and the *Listeria* lag are assumed (Appendix A). The spread in minimum temperature is the most uncertain: the thesis reports a 95% interval of 0.7 K on the mean, but the spread between sources is 2.8 K, and the value used here (1.5 K for spoilage flora) lies between them.

Shared structure. The reference and the device both turn the temperature history into something close to the time integral of the growth rate (Chapter 4). So the benchmark does not test whether a kinetic score beats a simple rule because it models growth better. It tests how much accuracy is lost to the sources of error listed. The structure-risk run relaxes one part of this by adding extra lag after shifts, but other structural failures, such as adaptation, population effects and species interactions, are not tested.

Constructed prevalence. About half of the spoilage trips and 8 to 38% of the *Listeria* trips are positive by construction. AUC does not depend on prevalence, but predictive values and calibration do, so calibration results show relative differences between methods and not absolute performance in the field.

Two organism classes. The main model stands in for psychrotrophic spoilage bacteria, and a case study covers *Listeria monocytogenes*. A review of dairy spoilage reports that about 40 to 50% of fluid-milk spoilage in the United States comes instead from spore-forming gram-positive bacteria, which grow later and differently [13]. Nothing here estimates illness risk, which depends on dose, virulence, the host and the contamination event.

Rates, not curves. Records are maximum rates from many sources and methods, not curves, so lag and responses to temperature shifts are not tested on them. The *Listeria* held-out test is by condition, not by study.

Assumed trip generator. Excursion frequency, duration and the independence of trips are assumptions. Real chains have correlated failures, such as one bad truck affecting many pouches.

One published validation experiment. The model reproduces one published experiment, with approximate values read from the text. No new plate counts have been measured.

Novelty not confirmed. The idea behind the model was built before (Chapter 3). A systematic search of indexed literature has not been done, so the statement that no source read quantifies each error source on a common footing is a statement about the sources read.

What comes next

Four pieces of work would turn this into a validated result, in order of effort. First, extend the ComBase data to more spoilage organisms and to full growth curves, not only rates. Second, calibrate the sensors against a reference thermometer and measure the thermal time constants of real packs; this needs only a stirred bath, a reference thermometer and a refrigerator. Third, measure real plate counts of spoilage flora in commercial pasteurised milk under constant and fluctuating temperature, and predict the fluctuating groups blind, in a food microbiology laboratory with a supervisor and only organisms already present in commercial milk. Fourth, run a field logging study. The code, tests and data are open so that anyone can repeat or extend it.

Appendix A: Parameter Provenance

Every number the simulation uses that did not come from a measurement or a source is marked **assumed**. The values are read directly from the project's parameter file.

Parameter	Value	Provenance
Tmin mean	-7.7	SOURCED: Neumeyer 1995 thesis Table 4.9 (-7.7 C, 68 growth curves, broth and milk). Literature mean 265.4 K = -7.75 C. Lin 2016 own data give -9.15 C
Tmin sd	1.5	ASSUMED. Neumeyer Table 5.1: +/-0.7 K is the 95% CI of the MEAN over 64 sources; their SD is 2.83 K but includes method noise. 1.5 K is a compromise to be replaced by a measured strain SD
b nominal	0.030	SOURCED, two independent sources agree: Neumeyer 1995 Table 4.9 (b=0.00465 on sqrt(1/generation time in min)) converts to 0.030; Lin et al. 2016 (growth rates 0.056/0.17/0.46 per h at 4/15/29 C, log10 basis) gives 0.027
b cv	0 / 0.10 / 0.25	ASSUMED sensitivity axis, not measured
h0 truth	U(2.5,5.0)	SOURCED RANGE from Lin 2016 Table 2: lag x growth rate (ln basis) = 3.8, 5.0 and 3.0 at 4, 15, 29 C. Inoculated UHT milk, so natural contamination may differ
h0 estimator	3.8	SOURCED: Lin 2016 at 4 C (lag 29.5 h)
log10N0 truth	U(1.5,2.0)	ASSUMED. Martin et al. 2021 notes post-pasteurisation contamination is often very low (1 to 50 cells per 100 mL), so this is probably high and needs measuring
log10N0 estimator	1.7	ASSUMED
log10Nmax	9.0	ASSUMED (typical order for milk psychrotrophs)
Nlimit A	3e4	SOURCED as a number: FSSAI Appendix B v2 (04.11.2024) Table 2A, pasteurized milk APC m=3x10^4/mL. A process-hygiene criterion used here as a quality marker. US PMO limit 20,000 CFU/mL (Martin 2021)
Nlimit B	1e6	SUPPORTED: about 10^6 CFU/mL is where quality deteriorates (Martin et al. 2021, citing Ranieri & Boor 2009; Lin 2016 citing Momani & Natsheh 2012)
label shelf life	83 (A) / 113 (B)	DESIGN: time for the estimator at nominal parameters held at 4 C to reach the limit; set for label balance before any method comparison
events per trip	Poisson, 1 per 144 h of horizon, max 8	ASSUMED
event peak ambient	12 / 14.3 / 17.5 / 30.7 (w .30/.25/.30/.15)	14.3, 17-18.3, 30.6-30.7 from the team's evidence deck (field spot readings and one study, to be re-verified); 12 ASSUMED
event duration	lognormal median 60 min, sigma 0.8, clip 10-480	ASSUMED
product time constant	low-mass U(5,20) min; insulated U(1,4) h	ASSUMED, to be measured (lab plan, part A)
base temp	U(2.5,5.0)	ASSUMED
shift lag fraction	0 (baseline) / 0.25 (structure-risk run)	SOURCED VALUE, ASSUMED IMPLEMENTATION: Zwietering et al. 1994 (additional lag is one quarter of the lag at the new temperature, Lactobacillus plantarum); Lin 2016 saw extra lag after shifts in milk
sensor DS18B20	bias U(-0.5,0.5), noise sd 0.05, res 0.0625	spec +/-0.5 C from the team's plan and datasheet; noise ASSUMED
sensor PT1000 MAX31865	bias U(-0.15,0.15), noise sd 0.03, res 0.03125	spec from the team's plan; noise ASSUMED
combase fit	b 0.0256, Tmin -6.6 C (n=50 robust); log resid sd 0.16	EMPIRICAL: 51 ComBase records (Tasmanian Institute of Agriculture, pasteurised milk, 2.7-14.9 C); one record with an implausible temperature removed; Neumeyer/Lin parameters overpredict these by 1.6x

Parameter	Value	Provenance
runs reported	v4 baseline; v4 extra lag; v5 low starting load; v6 ComBase-fitted	all use seeds fixed in src/benchmark_v2.py; 4000 trips per setting
lis b	0.0235	FITTED: 46 ComBase records of Listeria monocytogenes in milk, conditions not specified, 2-16 C; bootstrap 95% 0.0203-0.0265. Held-out check on 53 records: bias factor 1.02, accuracy factor 1.37 (split by condition, not by study)
lis Tmin	-2.3	FITTED: same fit; bootstrap 95% -4.2 to -0.9 C
lis Tmin sd	1.0	ASSUMED: of the order of the bootstrap standard error (about 0.8 K)
lis h0	3.0 (estimator); U(2.0,4.5) (truth)	ASSUMED: no Listeria lag data were extracted; borrowed from the Pseudomonas range
lis N0	U(-1.25,-0.75) narrow; estimator -1.0	ASSUMED: about 0.06-0.18 cells per mL. Kept narrow on purpose: the +0.5 log label sits at -0.5, so a wider start range would put some trips above the label at time zero
lis labels	+0.5 log10 (A) and +1 log10 (B) growth over starting load; horizons 100 h and 202 h	ASSUMED round-number labels, NOT regulatory limits (FSSAI Table 2B requires absence in 25 mL)

Appendix B: Reproducing the Results

The repository contains everything needed to regenerate the numbers, tables and figures in the paper and in this book from fixed random seeds.

Layout. `src/` holds the kernel, the simulator, the benchmark and the validation scripts; `tests/` holds the 14 tests; `data/` holds the parameter table ; `results/` holds the output of each benchmark run; `paper/`, `book/` and `demo/` hold the sources of the manuscript, this book and the interactive demo; `docs/` is the published website with the finished PDFs.

Commands. `make test` runs the tests. `make runs` repeats the five benchmark runs (about three hours on two cores). `make summary` condenses them, `make paper` and `make book` rebuild the documents, and `make all` does everything. Individual runs are selected by environment variables, for example `DG_PARAMS=lis python3 src/benchmark_v2.py 4000 results/lis`.

Runs reported. (a) Baseline with sourced biology; (b) the same with extra lag after temperature shifts; (c) a sensitivity run with a realistic low starting load; (d) a run with the growth parameters fitted to the ComBase records of natural spoilage flora; (e) a run for *Listeria monocytogenes* with parameters fitted to ComBase records. An earlier run with placeholder parameters showed the same overall pattern and is not reported.

Licence and archive. The code is released under the MIT licence, with a citation file; a persistent archive identifier can be added when a version is deposited.

References

Only sources that were actually opened are listed. Where only an abstract or a registry entry was read, the entry says so.

[1] FSSAI, *Appendix B: Microbiological Requirements*, Version 2 (04.11.2024), Tables 2A and 2B (milk and milk products). Food Safety and Standards Authority of India.

[2] K. Neumeyer, *Modelling pseudomonad growth in milk and milk-based products*, M.Sc. thesis, University of Tasmania (repository record eprints.utas.edu.au/21013).

[3] K. Neumeyer, T. Ross and T. A. McMeekin, "Development of a predictive model to describe the effects of temperature and water activity on the growth of spoilage pseudomonads," *Int. J. Food Microbiol.*, vol. 38, pp. 45–54, 1997.

[4] K. Neumeyer, T. Ross, G. Thomson and T. A. McMeekin, "Validation of a model describing the effects of temperature and water activity on the growth of psychrotrophic pseudomonads," *Int. J. Food Microbiol.*, vol. 38, pp. 55–63, 1997.

[5] H. Lin, M. Shavezipur, A. Yousef and F. Maleky, "Prediction of growth of *Pseudomonas fluorescens* in milk during storage under fluctuating temperature," *J. Dairy Sci.*, vol. 99, pp. 1822–1830, 2016.

[6] F. Tarlak, "The use of predictive microbiology for the prediction of the shelf life of food products," *Foods*, vol. 12, no. 24, art. 4461, 2023, doi:10.3390/foods12244461.

[7] E. Stavropoulou and E. Bezirtzoglou, "Predictive modeling of microbial behavior in food," *Foods*, vol. 8, art. 654, 2019.

[8] V. Kumar, J. J. Ahire and N. K. Taneja, "Advancing microbial food safety and hazard analysis through predictive mathematical modeling," *The Microbe*, vol. 2, art. 100049, 2024.

[9] O. R. Taiwo, H. Onyeaka, E. K. Oladipo, J. K. Oloke and D. C. Chukwugozie, "Advancements in predictive microbiology: integrating new technologies for efficient food safety models," *Int. J. Microbiol.*, vol. 2024, art. 6612162, 2024, doi:10.1155/2024/6612162.

[10] I. J. Opara *et al.*, "AI-driven predictive microbiology with real-time sensors for next-generation food safety," *J. Life Sci. Public Health*, vol. 1, no. 2, 2025, doi:10.69739/jlsph.v1i2.1299.

[11] L. Lima, A. I. Pereira, M. Pintado and M. Carocho, "Fresh dairy quality in the era of smart food systems: deterioration pathways, analytical technologies, and predictive modeling," *Foods*, vol. 15, no. 19, art. 3439, 2026, doi:10.3390/foods15193439.

[12] C. N. Ugwu *et al.*, "Digital twins for food processing and shelf-life: linking mechanistic models, sensors, and AI to predict quality, safety, and waste through near-real-time decision support," *Front. Food Sci. Technol.*, 2026, doi:10.3389/frfst.2026.1813819.

[13] N. H. Martin, P. Torres-Frenzel and M. Wiedmann, "Invited review: Controlling dairy product spoilage to reduce food loss and waste," *J. Dairy Sci.*, vol. 104, pp. 1251–1261, 2021, doi:10.3168/jds.2020-19130.

[14] J. Baranyi and T. A. Roberts, "A dynamic approach to predicting bacterial growth in food," *Int. J. Food Microbiol.*, vol. 23, pp. 277–294, 1994.

[15] S. Tsaloumi, Z. Aspridou, E. Spyrelli, G.-J. E. Nychas and K. Koutsoumanis, "Development and validation of a mathematical model for pseudomonads growth as a basis for predicting spoilage of fresh poultry breast and thigh fillets," *Poultry Sci.*, vol. 101, art. 101985, 2022, doi:10.1016/j.psj.2022.101985.

[16] M. H. Zwietering, J. C. de Wit, H. G. A. M. Cuppers and K. van 't Riet, "Modeling of bacterial growth with shifts in temperature," *Appl. Environ. Microbiol.*, pp. 204–213, Jan. 1994.

[17] World Health Organization, "WHO's first ever global estimates of foodborne diseases," news release, 3 Dec. 2015.

[18] ComBase (combase.cc), growth records in milk: aerobic spoilage bacteria in pasteurised milk (Tasmanian Institute of Agriculture; 51 records) and *Listeria monocytogenes* (99 records with a stated rate, 2-16 °C), accessed 19 Oct. 2026.

[19] A. Sofat and B. Sodhi, "Beyond thresholds: a quality-aware decision intelligence framework for cold chain IoT systems," arXiv:2608.15082, 2026.

[20] S. Lau, A. Trmčić, N. H. Martin, M. Wiedmann and S. I. Murphy, "Development of a Monte Carlo simulation model to predict pasteurized fluid milk spoilage due to post-pasteurization contamination with gram-negative bacteria," *J. Dairy Sci.*, vol. 105, pp. 1978–1998, 2022, doi:10.3168/jds.2021-21316.

[21] A. J. Buehler, N. H. Martin, K. J. Boor and M. Wiedmann, "Psychrotolerant spore-former growth characterization for the development of a dairy spoilage predictive model," *J. Dairy Sci.*, vol. 101, pp. 6964–6981, 2018, doi:10.3168/jds.2018-14501.

[22] C. Qian, S. I. Murphy, T. T. Lott, N. H. Martin and M. Wiedmann, "Development and deployment of a supply-chain digital tool to predict fluid-milk spoilage due to psychrotolerant sporeformers," *J. Dairy Sci.*, vol. 106, pp. 8414–8433, 2023, doi:10.3168/jds.2023-23673.

[23] M. C. Giannakourou and P. S. Taoukis, “Holistic approach to the uncertainty in shelf life prediction of frozen foods at dynamic cold chain conditions,” *Foods*, vol. 9, no. 6, art. 714, 2020, doi:10.3390/foods9060714.

[24] N. Ndraha, H.-I. Hsiao, J. Vlajic, M.-F. Yang and H.-T. V. Lin, “Time-temperature abuse in the food cold chain: review of issues, challenges, and recommendations,” *Food Control*, vol. 89, pp. 12–21, 2018.